Suggested problems: 1-13 odd, 17-29 odd

Problem 1: The price P in dollars to purchase a used car is a function of its original cost, C, in dollars, and its age, A, in years.

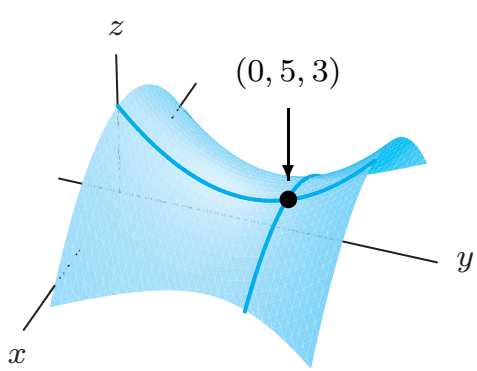
(a) What are the units of $\partial P/\partial A$?

(b) What is the sign of $\partial P/\partial A$ and why?

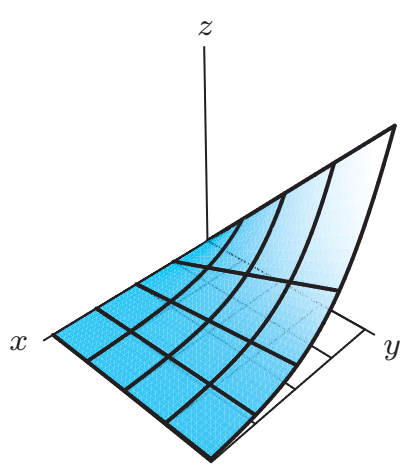
(c) What are the units of $\partial P/\partial C$?

(d) What is the sign of $\partial P/\partial C$ and why?

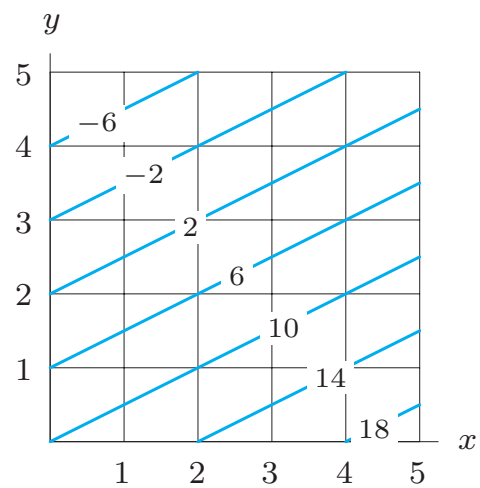
Problem 2: The figure below shows the surface $z = f(x, y)$. What is the sign of $f_x(0, 5)$? What is the sign of $f_y(0, 5)$?



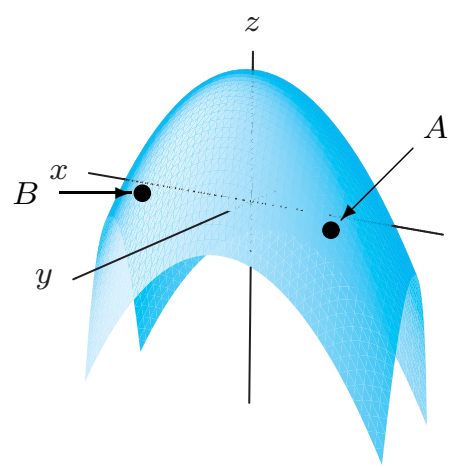
Problem 3: The figure below shows the graph of a function $x = f(x, y)$ on the domain $0 \le x \le 4$ and $0 \le y \le 4$. Use the graph to rank the following quantities in order from smallest to largest: $f_x(3, 2)$, $f_x(1, 2)$, $f_y(3, 2)$, $f_y(1, 2)$, 0.



Problem 4: Suppose we are given the contour diagram below of a function $z = f(x, y)$. Is f_x positive or negative? How about f_y?



Problem 5: The graph of $z = f(x, y)$ is shown in the figure below. The points A and B are in the xy-plane.

(a) What is the sign of $f_x(A)$? What is the sign of $f_y(A)$?

(b) The point P in the xy-plane moves along a straight line from A to B. How does the sign of $f_x(P)$ change? How does the sign of $f_y(P)$ change?